

CHAPTER ONE

What is *Statistics?*

Why business needs data, what kinds of data exist, and how to use numbers honestly.

3

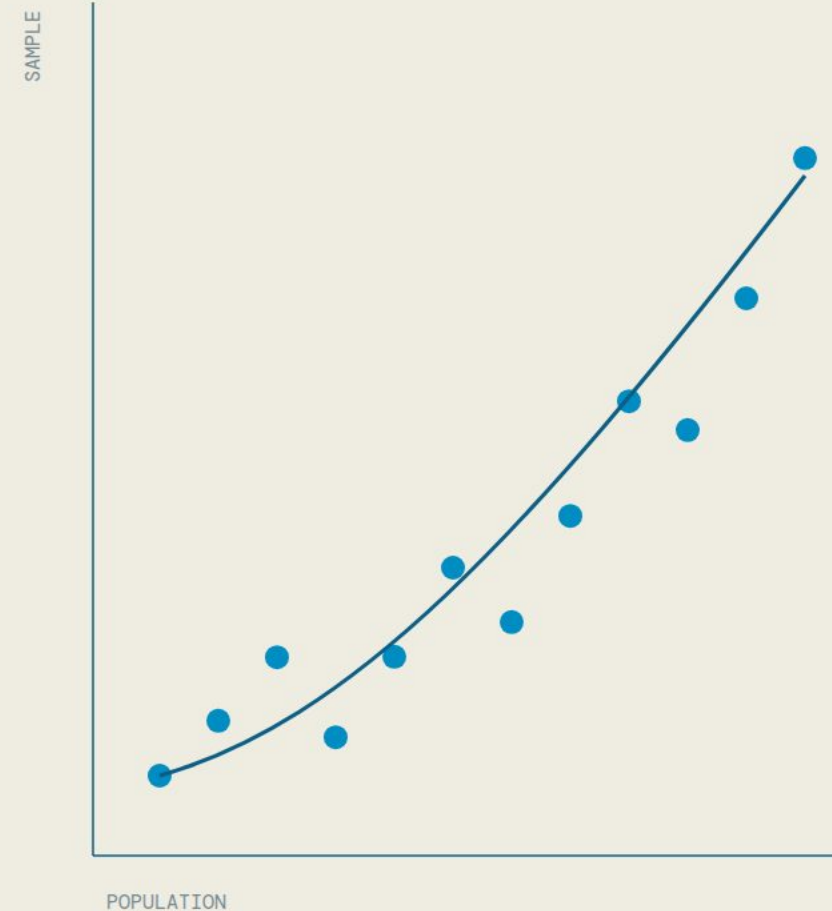
LEARNING OBJECTIVES

32

SLIDES

22

PRACTICE POOLS



Three learning objectives

CHAPTER 1 • CONTENTS

1.1

UNDERSTAND STATISTICS AND DATA BASICS

Why statistics matters for business decisions. Statistics is the science of collecting, organizing, analyzing, interpreting, and presenting data.

1.2

DIFFERENTIATE AND CLASSIFY DATA

Population vs. sample, descriptive vs. inferential statistics, qualitative vs. quantitative variables, and the four levels of measurement.

1.3

RECOGNIZE ETHICAL IMPLICATIONS

Honesty, transparency, and objectivity when collecting, analysing, and reporting data.

Everything you meet here — population, sample, parameter, statistic, variable, level — returns in every later chapter. Two practice games and 22 practice pools support this chapter.

1.1

LEARNING OBJECTIVE

Understand statistics *and* data basics

Why is statistics important for business decisions? What exactly is statistics? Where do you meet it — at work and in your own life?

CONTENTS • 8 SLIDES

- 01 The world is built on data
- 02 Three core reasons
- 03 The value of statistics
- 04 Business applications
- 05 Example: stc
- 06 Personal decisions
- 07 The definition • five steps
- 08 Check your understanding

The world is built on data

LO 1.1 • Understand statistics and data basics

Every day, in Saudi Arabia and everywhere, we create data. Think about it:

01 Shopping

Every purchase you make online or in a store like Panda or Jarir Bookstore is recorded.

02 Government apps

Every time you use Absher or Tawakkalna, data is used.

03 Social media

Every post, “like”, or view on X, TikTok, or YouTube adds to a huge pool of data.

Raw data by itself is not very useful. It is like millions of puzzle pieces that are not connected. The real power comes when we understand the data and use it to make smart decisions — **this is where statistics comes in.** Knowledge of statistics is no longer only for scientists. It is a basic skill for almost every career: business, finance, marketing, and government. Saudi **Vision 2030** asks for efficiency, data-driven government, and a diversified economy — so the demand for people who can “speak the language of data” is higher than ever.

IN ONE SENTENCE

Data everywhere → statistics turns it into decisions.

Three core reasons

LO 1.1 • Understand statistics and data basics

01

Data is everywhere — and it needs statistics to be useful

The amount of data collected today is huge (“Big Data”): millions of gigabytes every minute. A list of 10,000 sales transactions is only raw data; a manager cannot learn anything from it at a glance. Statistics gives us the methods to organize, summarize, and present it in a useful way.

02

Statistical techniques help us make better decisions

The main goal of statistics in business is to move from guessing to informed decisions. In your career and in your personal life, statistics helps you judge situations and predict results.

03

Statistical literacy is a critical career skill

No matter your major — accounting, management, marketing, or finance — you will need a working knowledge of statistics to do well and to move up in your career.

The value of statistical knowledge

LO 1.1 • Understand statistics and data basics

“In God we trust; all others must bring data.”

W. EDWARDS DEMING • QUALITY-CONTROL PIONEER

When you make a claim at work, bring the numbers.

01 Critical thinking

Check facts, spot mistakes in reasoning, and tell the difference between two things that happen together and one thing that causes another.

02 Data-driven decisions

Business leaders make choices based on facts and data, not only on feelings — fewer personal biases, better plans.

03 Risk management

Measure uncertainty, estimate what may happen, and build strong plans to handle risk.

A data-driven business world

LO 1.1 · Understand statistics and data basics

	SKILL	WHAT IT GIVES YOU
01	Future-proof	AI and big data are built on statistics.
02	Smarter choices	Turn gut feelings into evidence-based strategies.
03	Understand markets	Analyze trends, customer behavior, and competition.
04	Optimize operations	Improve efficiency, reduce costs, raise quality.
05	Master marketing	Measure campaign success and return on investment (ROI).
06	Navigate finance	Assess risk, forecast trends, evaluate investments.

For business students in Saudi Arabia this is vital. As the Kingdom works toward Vision 2030, the economy is moving to knowledge-based industries that need analytical skills. Statistics is the bridge from raw data to business decisions — and every KSU Bachelor program requires it.

Professional applications

LO 1.1 • Understand statistics and data basics

01 Finance & accounting

An analyst at Al Rajhi Bank or SNB studies the past performance of stocks — the average return and the risk (standard deviation) — to advise clients.

02 Marketing & sales

A marketing manager at Almarai shows two new advertisements to two different samples of consumers and uses statistics to decide if there is a real difference in response.

03 Operations management

Saudia wants better on-time performance: analyze flight data, find the most common causes of delay, fix them.

04 Human resources

Workforce planning, salary analysis, performance evaluation, employee satisfaction surveys, better recruiting.

05 Strategic management

Competitor analysis, planning, scenario modeling, and tracking key performance indicators (KPIs).

06 Fast-growing economies

Statistics gives businesses in emerging markets the tools to understand complex markets and find growth opportunities.

Example: a manager at stc

LO 1.1 · Understand statistics and data basics

Imagine you are a manager at **stc**. You have data on the monthly data usage of **millions** of customers.

WITHOUT STATISTICS

RAW DATA · MILLIONS OF ROWS

41	86	80	27	58	88	71	91	85	19
88	12	71	44	81	40	35	71	80	81
71	61	92	30	40	92	30	77	60	12
96	19	31	86	16	49	14	45	71	87
60	65	61	84	67	28	57	23	15	28
74	38	44	97	66	91	49	64	75	60
84	55	79	85	63	85	40	54	98	14
46	88	96	31	52	80	84	83	24	94

WITH STATISTICS

INFORMATION

38 GB

50 GB

average usage per month

most popular package

USAGE BY MONTH · PEAKS IN RAMADAN AND SUMMER



→ decisions: packages · network investment · promotions

A massive, unreadable database becomes: the average usage, the most popular package, a chart of usage during the year, a comparison of regions — **information that helps you decide.**

Statistics in your personal decisions

LO 1.1 • Understand statistics and data basics

01 Buying a car

You worry about fuel cost. Compare the average kilometers per liter — a statistical measure — for different models and make a smart choice.

02 Reading the news

You read a news report that claims a new health product is effective. A knowledge of statistics allows you to question the study: How large was the sample? Was it a representative sample? Is the reported effect statistically significant?

STATISTICAL LITERACY PROTECTS YOU

It helps you judge claims in advertisements, news reports, and social media posts — misinformation is common in today's digital world. With basic statistical concepts you can:

- spot misleading information
- make better personal financial decisions
- understand health and medical information
- judge how reliable a news source is

DEFINITION

Statistics is the science of *collecting, organizing, analyzing, interpreting, and presenting* data.

It helps us find patterns, trends, and useful insights. In business it is the key tool that turns raw data into information for decisions. Karl Pearson called statistics “the grammar of science”: a clear, systematic way to describe uncertain things, test what we think, and confirm our findings.



In summary

LO 1.1 · Understand statistics and data basics

TAKE-AWAY

01	Organize data	Turn raw numbers into meaningful information.
02	Make decisions	Use statistical analysis to guide professional and personal choices.
03	Advance your career	Stand out in the job market with analytical skills.
04	Support Vision 2030	Contribute to Saudi Arabia's data-driven future.

Statistics gives you the tools to make sense of the huge amount of data in our world: turn raw data into useful information, use it to make better decisions, and become a more valuable professional in any field. This course is your first step.

Check your understanding

LO 1.1 • Understand statistics and data basics

- Q1** A manager at Jarir has a list of 20,000 sales receipts from last month. Which step of statistics turns this list into “average sale per customer = 245 SAR”? **A** Collecting **B** Analyzing **C** Presenting
- Q2** Which is the best example of using statistics to make a *business* decision? **A** Choosing the color of your car because you like it **B** Almarai showing two ads to two sample groups and comparing the response **C** Reading a list of 10,000 transactions one by one
- Q3** Complete the definition: Statistics is the science of _____, organizing, analyzing, _____, and presenting data.

ANSWERS

1 B — analyzing. **2** B. **3** collecting / interpreting.

1.2

LEARNING OBJECTIVE

Differentiate *and* classify data

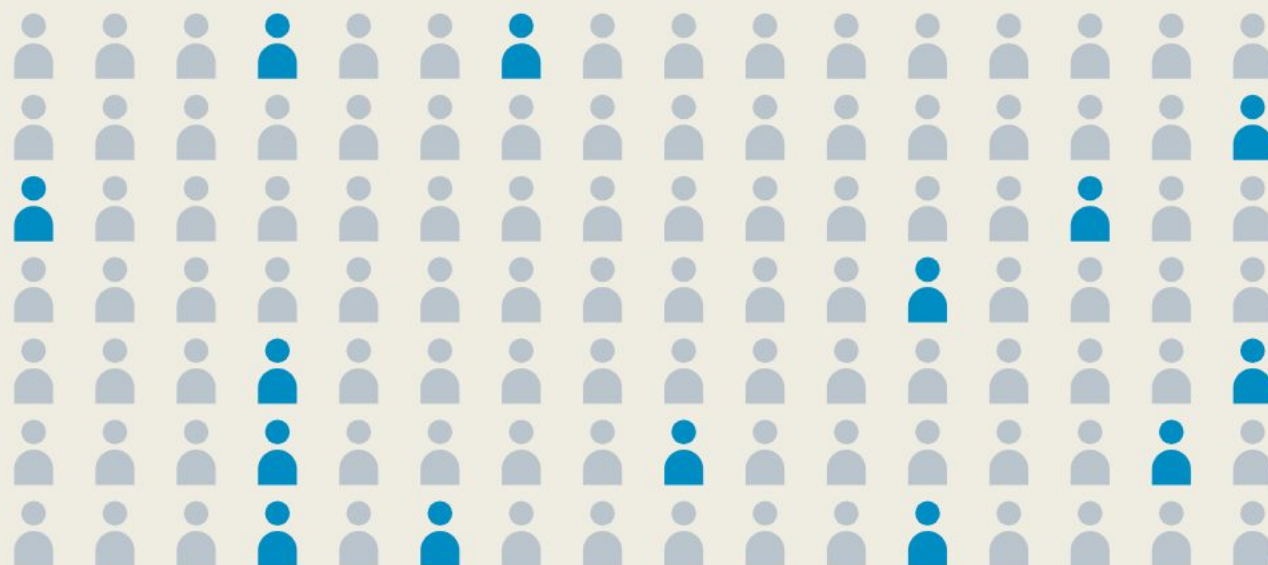
Population or sample? Describing data, or estimating from it? A word or a number — counted or measured? And which level of measurement?

CONTENTS • 8 SLIDES

- 01 Population and sample
- 02 Descriptive vs. inferential
- 03 What each one includes
- 04 Real-life examples
- 05 Key differences • working together
- 06 Types of variables
- 07 Levels of measurement (N-O-I-R)
- 08 Check your understanding

Population *vs.* sample

LO 1.2 · Differentiate and classify data



● the 500 employees who were surveyed — the sample ● the rest of the population

Each figure stands for many employees; the population is all 16,586 stc employees.

Why sample? Too big · too slow · too expensive · sometimes destructive — you cannot taste every Almarai bottle.

POPULATION

The **whole group** you want to learn about. For example, if we want to learn about stc employees, all 16,586 employees are the population. Complete, but usually too big, too slow, or too expensive to study fully. A number that describes the population is a **parameter**, written μ . Usually unknown.

SAMPLE

The **part** of the population we actually study. For example, 500 stc employees chosen to represent all of them. Cheaper and manageable. A number that describes the sample is a **statistic**, written $\bar{x}$. We calculate it.

RULE 1

Population →
Parameter

RULE 2

Sample →
Statistic

Descriptive *vs.* inferential

LO 1.2 · Differentiate and classify data

DESCRIPTIVE STATISTICS

Methods for **organizing, summarizing, and presenting** data in an informative way. They describe the data you have and make no conclusion beyond it.

INFERENCEAL STATISTICS

Methods that use data from a **sample** to make conclusions, estimates, or predictions about a **population**. Because we use only a sample, there is always some uncertainty. Signal words: *estimate, predict, conclude that all...*

What each one includes

LO 1.2 · Differentiate and classify data

DESCRIPTIVE

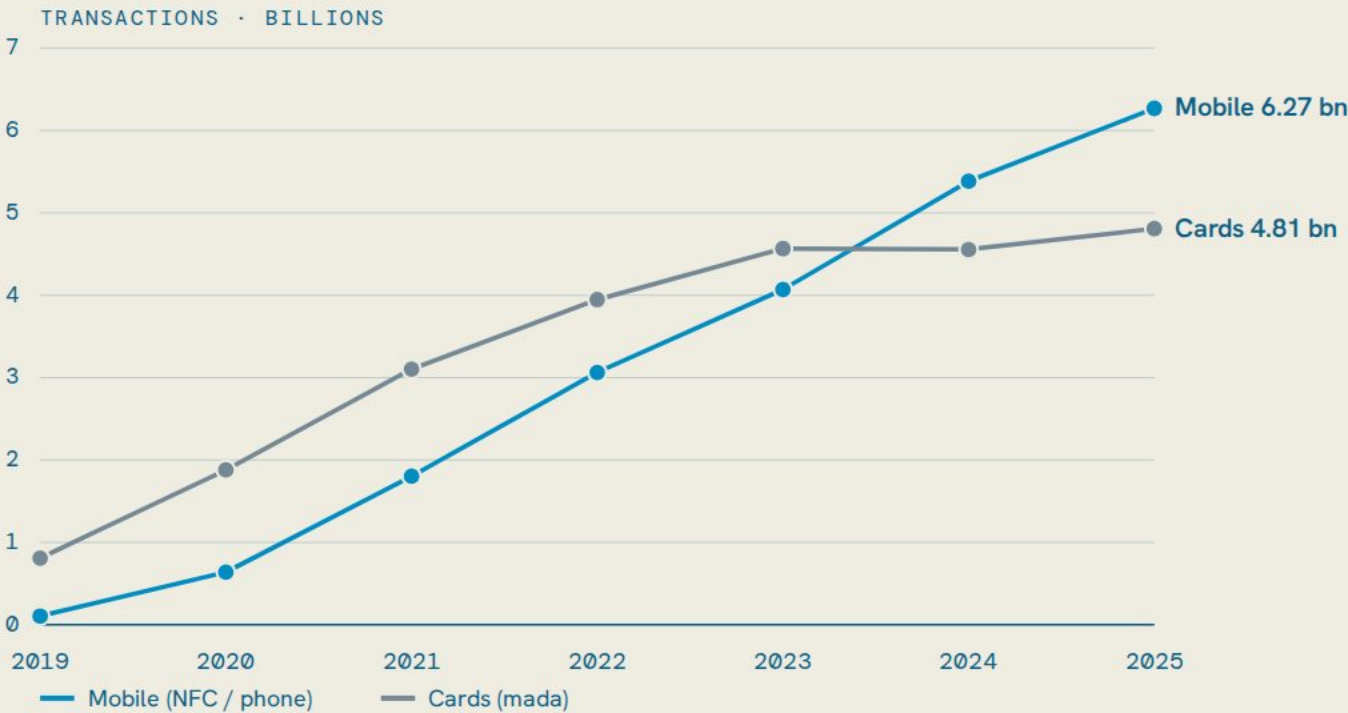
	COMPONENT	BUSINESS APPLICATION
01	Central tendency	Mean, median, mode · sales performance
02	Dispersion	Range, variance, standard deviation · customer demographics
03	Visualization	Histograms, box plots, scatter plots · inventory turnover
04	Distributions	The shape of the data · employee metrics

INFERENTIAL

	COMPONENT	BUSINESS APPLICATION
01	Probability & sampling	A sample that represents the population · market research
02	Hypothesis testing	Test a claim about a population · quality control
03	Regression analysis	Model relationships to predict · financial forecasting
04	Statistical modeling	Forecast results, find trends · A/B testing

Mobile overtook cards *in 2024*

SAMA · MADA POS STATISTICS · 2019-2025



MOBILE POS TRANSACTIONS · 2025

6.27_{bn}

up from 0.11 bn in 2019. Card transactions grew to 4.6 bn by 2023 and have stayed near that level (4.81 bn in 2025).

WHY THIS IS DESCRIPTIVE

The numbers only **summarise what happened**: two totals per year, one line each. No sample, no estimate about a larger group.

ALSO DESCRIPTIVE

Saudi Exchange 2025: SAR 1,299 bn traded over 251 days ≈ **SAR 5.2 bn per day**, 30% below 2024.

Inferential statistics *in real life*

SAMA · KEY ECONOMIC DEVELOPMENTS · Q1 2026

AVERAGE CPI · YEAR-ON-YEAR · Q1 2026

1.8%

The Saudi Central Bank's *Key Economic Developments* report for Q1 2026 shows the average Consumer Price Index up 1.8% year-on-year (0.4% quarter-on-quarter), a moderation from the 2.1% peak observed in Q1 2025 — driven mainly by personal care and miscellaneous services (+8.1%) and housing, water, electricity, gas and other fuels (+4.1%).

Source: SAMA, Key Economic Developments – Q1 2026 (prices collected by GASTAT).

WHY IS THIS INFERENTIAL?

Nobody records every price in the Kingdom. GASTAT prices a fixed basket of 582 items in a **sample** of shops and cities and uses it to estimate the change in prices for the whole **population** of consumers. SAMA and companies then use this estimate to decide on interest rates, salaries, and prices.

A CUSTOMER SURVEY

A company surveys 1,000 customers: 62% like the new package. It estimates that between 59% and 65% of **all** customers like it — the range shows the uncertainty of using a sample.

The key differences

LO 1.2 · Differentiate and classify data

	DESCRIPTIVE STATISTICS	INFERENTIAL STATISTICS
Main goal	Organize, summarize, and present the data you have.	Use data from a sample to make an estimate or decision about a population.
Data used	Data from the whole group of interest (e.g., all students in one class).	Data from a sample taken from a larger population.
Conclusion	A fact about the data you analyzed.	An educated guess (estimate) about the population — there is always a chance of being wrong.
Example	“The average height of the 30 players on the Al-Nassr football team is 181 cm.”	“Based on a sample of 100 players, we estimate the average height of all players in the Saudi Pro League is between 179 cm and 183 cm.”

They work together, in order: a question about a **population** → collect a **sample** → **describe** the sample (averages, charts) → **infer** a conclusion about the population. Chapters 2–4 are descriptive; Chapters 5–7 build probability; inference is QUA 207.

Classifying variables

LO 1.2 · Differentiate and classify data

Qualitative (categorical)

The value is a **category or a label**, not a measured number.

- Examples: gender, nationality, marital status, product category, satisfaction level (poor / fair / good / excellent).
- Business use: market segmentation, customer profiling, quality classification.

Quantitative (numerical)

The value is a **number** we can measure or count.

- Examples: sales revenue, number of employees, production time, customer age, inventory level.
- Business use: financial analysis, productivity measurement, demographic studies.

The type of variable is the first step in choosing the right statistical method: you can average ages, but you cannot average nationalities. Numbers used only as labels — employee ID, postal code — are qualitative.

Discrete *or* continuous?

LO 1.2 · Differentiate and classify data

Discrete

Only certain separate values, usually whole numbers from **counting**.

- Number of employees, customer complaints, defective items in a batch, store locations.
- Business use: inventory management, staffing decisions, quality control.

Continuous

Any value in a range, usually from **measuring**.

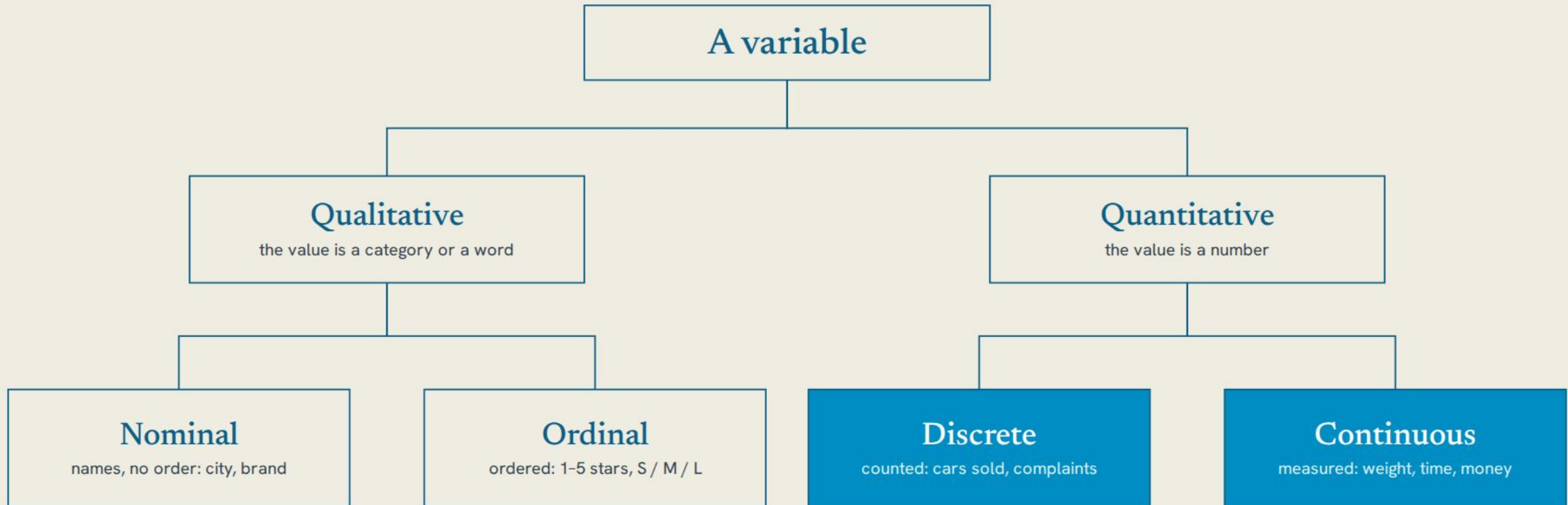
- Time to finish a task, product weight, temperature, customer spending.
- Business use: process improvement, financial modeling, resource allocation.

COURSE CONVENTION

Counts are discrete. Measured amounts are continuous.

Summary of variable types

LO 1.2 · Differentiate and classify data



Ask: word or number? If a number: do I count it or measure it?

Levels of measurement

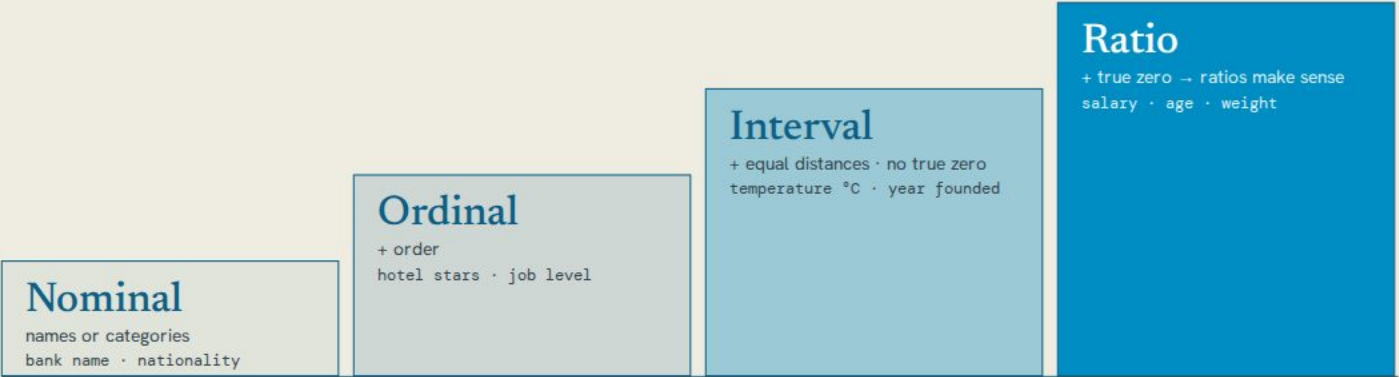
LO 1.2 · Differentiate and classify data

LEVEL	WHAT IT MEANS	OPERATIONS	EXAMPLES
01 Nominal	Names or categories with no natural order. We can only count how many are in each category.	= / ≠	gender · product category · nationality
02 Ordinal	Data can be ranked, but the distance between ranks is not equal.	< / >	satisfaction poor-fair-good-excellent · education level
03 Interval	Numbers with equal distances but no true zero (zero does not mean “nothing”), so ratios have no meaning.	+ / −	temperature °C · IQ · calendar years
04 Ratio	Equal distances and a true zero, so ratios make sense: 20 SAR is twice 10 SAR.	× / ÷	revenue · age · weight · market share

Why is °C interval and not ratio? 0 °C does not mean “no heat”. 20 °C and 40 °C are 68 °F and 104 °F — and 104 is not twice 68. Money is ratio: 40 SAR is twice 20 SAR in any currency.

The N·O·I·R ladder

LO 1.2 · Differentiate and classify data



LEVEL	CATEGORIES?	ORDER?	EQUAL DISTANCES?	TRUE ZERO?	BUSINESS EXAMPLE
Nominal	yes	—	—	—	Bank name (Al Rajhi, SNB, Riyad)
Ordinal	yes	yes	—	—	Hotel rating (1★ to 5★)
Interval	yes	yes	yes	—	Temperature in °C, year founded
Ratio	yes	yes	yes	yes	Salary in SAR, number of employees

Check your understanding

LO 1.2 · Differentiate and classify data

- Q1** A researcher surveys 400 KSU students: average daily study time 3.2 hours. She concludes that the average for *all* KSU students is between 3.0 and 3.4 hours. The number 3.2 is a _____ and the conclusion is _____ statistics. **A** parameter; descriptive **B** statistic; inferential **C** statistic; descriptive
- Q2** Classify each variable: (a) number of cars sold by a dealership last month, (b) time to serve a customer at Al Baik, (c) type of payment (cash / mada / Apple Pay), (d) customer rating (very unhappy ... very happy).
- Q3** Which level of measurement is “temperature in Riyadh in °C”? Why is it not ratio?

ANSWERS

1 B. 2 (a) quantitative discrete · (b) quantitative continuous · (c) qualitative nominal · (d) qualitative ordinal. **3** Interval — 0 °C does not mean “no temperature”, so 40 °C is not “twice as hot” as 20 °C.

1.3

CONTENTS • 4 SLIDES

- 01 Six ethical values
- 02 Charts and numbers that mislead
- 03 Chapter summary
- 04 Check your understanding

LEARNING OBJECTIVE

Recognize *ethical* implications

Numbers can inform — or mislead. The six values of honest statistical work, and the tricks you must recognise and never use.

Six ethical values

LO 1.3 • Recognize ethical implications

01 Objectivity

Analyze without a “wanted” answer in mind. Let the data lead. Never “cherry-pick” only the data that supports your story.

02 Transparency

Write down what you did: data source, cleaning, methods, assumptions, limitations — so others can check and repeat your work.

03 Accuracy

Use the correct method and check the results. Report the margin of error and possible bias. No false precision.

04 Responsibility

You are responsible for how your results are used. Explain the limits. Speak up when someone misuses statistical information.

05 Confidentiality

Respect privacy; anonymize and secure the data. In Saudi Arabia the Personal Data Protection Law (PDPL) makes this a legal duty.

06 Professional integrity

Keep professional standards, give credit to others, follow the ethical code — and refuse misleading practices even under pressure.

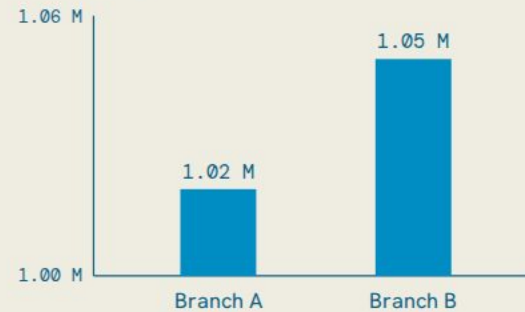
Charts and numbers that mislead

LO 1.3 • Recognize ethical implications

Honest: the axis starts at 0



Misleading: the axis starts at 1.00 M



Both charts show the same two profits. Only the second one “lies” — the values are written on the bars, never on the axis.

- 01 **Cherry-picked period**
“Sales grew 40%!” — measured from the worst month.
- 02 **Average without spread**
“Average salary 15,000 SAR” hides that most earn 6,000 and one manager 200,000.
- 03 **Unrepresentative sample**
Asking your friends on X what “everyone” thinks.
- 04 **Correlation as cause**
Ice-cream sales and electricity bills both rise in summer — one does not cause the other.

Chapter 1 in four ideas

SUMMARY

01 Statistics is essential for business success

In a data-driven world, statistical literacy is a practical need for good decisions in every department — not only an academic subject.

02 Descriptive and inferential methods serve different purposes

Descriptive statistics summarize the data you have; inferential statistics use a sample to estimate and test claims about a population.

03 Variable classification guides the analysis

Qualitative or quantitative, discrete or continuous, and the level of measurement (N-O-I-R) decide which methods are valid.

04 Ethical values protect the integrity of statistics

Objectivity, transparency, accuracy, responsibility, confidentiality, and professional integrity make statistical work trustworthy.

Check your understanding

LO 1.3 • Recognize ethical implications

- Q1** A marketing analyst runs the same test five times and reports only the one run that shows the campaign worked. Which ethical value is broken? **A** Confidentiality **B** Objectivity **C** Accuracy
- Q2** A company report says “customer satisfaction is 87.4638%” based on a survey of 20 people. What is the problem?
- Q3** Your manager asks you to share a customer list with names and phone numbers with a friend's company. What should you do, and which value applies?

ANSWERS

1 B. **2** False precision — a sample of 20 cannot support four decimal places (accuracy). **3** Refuse and explain — confidentiality and professional integrity; it may also break the PDPL.

CHAPTER 1 • PRACTICE

Play, then *practise.*

NEXT

Chapter 2 – Describing
Data: Frequency Tables,
Frequency Distributions,
and Graphic Presentation

[← Back to the QUA 107 course page](#)

- 01

Variable Sorter

Drag variables into the right box and choose the level of measurement.

↗
- 02

Stats Detective

Population or sample? Descriptive or inferential? Which ethical value?

↗
- 03

TheKey practice pools

CH1LO1 to CH1LO3 • 22 practice pools with many versions of each question.

↗